



ELECT MAINT Department
SIMHADRI

Note No :1

Ref:SMTTPP/ELECT MAINT/2024-25/LMI /588412

Date:

15/10/2024(17:26:32)

Subject:Revision of LMIOGNOPSELEC010 MAINTENANCE AND TESTING
OF GENERATOR EXCITATION CONTROL SYSTEMS

**Revision of LMI/OGN/OPS/ELEC/010, MAINTENANCE AND TESTING OF GENERATOR
EXCITATION CONTROL SYSTEMS**

LMIs are reviewed periodically for further improvement and incorporation of Recommendation by internal or external audit committee and COS.

OGN/OPS/ELEC/010, has been reviewed by COS.

Records of revision: NIL

LMI is reviewed.

Put up for kind approval.

Attachments:

1. LMI/OGN/OPS/ELEC/010, Rev:00, OCT 2024
2. OGN/OPS/ELEC/010

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NTPC

SIMHADRI

LOCATION MANAGEMENT INSTRUCTION

**MAINTENANCE AND TESTING OF
GENERATOR EXCITATION CONTROL SYSTEMS**

LMI/OGN/OPS/ELEC/010 Rev:00

OCT 2024

Approved for Implementation by HOP (SIMHADRI)

LMI APPROVAL DOCUMENT

1	LMI NAME	MAINTENANCE AND TESTING OF GENERATOR EXCITATION CONTROL SYSTEMS
2	LMI No.	LMI/OGN/OPS/ELEC/010 Rev:00
3	Reference Document	OGN/OPS/ELEC/010
4	Superseded Document	LMI/OGN/OPS/ELEC/010 Rev:00 Aug 22
5	Prepared By	Name: Sumit Kumar Verma S M (EMD)
6	Checked By	Name: Balabalaji Akula, DGM (EMD)
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8	Reviewed By	Name: P K Jena G M (O&M)
	Approved By	Name: S.K.Sinha H O P (Simhadri)

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1.0 Introduction

A fully functional, Automatic Voltage Regulator System (AVR) is necessary to allow safe, flexible, operation of a generator within its loading capability; to keep the generator voltage within specific limits at any steady load; to quickly restore the stator voltage following any sudden change in Unit loading or System Fault, and to assist in maintaining the stability of the Grid/Distribution System. A suitable level of Testing and Maintenance is necessary to ensure the reliability of AVR system performance and to prove compliance with Grid Code requirements.

2.0 Superseded Documents

LMI/OGN/OPS/ELEC/010 Rev:00 Aug 22

3.0 Details of Revision

NIL

4.1 METHODOLOGY OF IMPLEMENTATION

Testing and maintenance work would be carried out on both auto channels of DAVR & excitation control system, since both are integral part of the Generator Excitation control system.

4.2 MAINTENANCE & FUNCTIONAL CHECKS

Maintenance and Functional checks during shutdown include the following:

1. Physical inspection of all components, cabling, terminations, and Relay contacts.
2. Functional checking (Close/Trip) of Field circuit breaker
3. Insulation tests.
4. Maintenance of all mechanical moving parts.
5. AVR channel changeover (after voltage build up and AVR trouble to GRP trip signal isolated) on opportunity (not less than three months)

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4.3 STATIC TESTS

Static Testing includes the following and to be carried out when Generator is offline

1. Obtaining the characteristics of the control stages of the ECS – both automatic channels by secondary injection and optimizing these results compared to the desired characteristics.
2. Ensuring that all the ECS protection and monitoring devices are functional and set.
3. Verification of tightness of each control/power cable/ connectors termination to electronic modules, transducers, control transformers and TBs.
4. PMG fuse and control fuse bases should be secured properly and checked for terminal looseness. PMG fuses shall be replaced in every eight years.
5. Sealing of panel/ door/ spare cable glands.
6. Electronic modules are to be inspected for dust build-up, any heating marks on components, proper jumper setting/ parameter setting as per OEM manual/ instruction. Cubicle filter cleaning is to be done.
7. Inspection and testing of Power thyristors and associated electronic circuitry.
8. Ensuring Electronic earthing continuity and proper grounding of electronic modules.
9. Checking of Insulation resistance value of AVR trouble to GRP trip control cables and Power cables/ bus duct. Physical inspection of cable routing near exciter for any damage.
10. Checking of auxiliary power supply changeover scheme for control supply to electronic module and cooling fans of thyristor bridge.
11. Checking of thyristor cooling blowing motor vibrations and conditions. Power supply sources cooling fan blowers shall be checked for proper tightness of terminations, cable condition, fuse healthiness.
12. Testing of control and instrument transformers of AVR panel.
13. All jumper connections should be checked as per Factory test report.
14. List of parameters for separate functional groups would be checked and ensured as per Factory test report.
15. All DC and AC supply inputs to DAVR would be checked.
16. Output of all transformers would be checked.
17. Annunciations signals to control room shall be checked and joint protocol with operation is to be prepared.
18. Calibration of Parameter – Ug: Inject 3 phase voltage to the PT input terminal of both

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channels of DAVR and measure the corresponding voltage levels at DAVR (Ref: Annexure-I)..

19. Calibration of Parameter – Ig: Inject 3 phase current to the CT input terminal of both channels of DAVR and measure the corresponding voltage levels at DAVR. Check the effectiveness of the CT shorting links (Ref: Annexure-II).
20. Calibration of Parameter – If: Inject 2 phase current to the field current input terminal of both channels of DAVR and measure the corresponding voltage levels at DAVR (Ref Annexure-III)
21. Light load test for checking the healthiness of the Thyristors (Details Ref – Annexure IV).
22. Firing Pulses at pulse generation unit/pulse intermediate stage/pulse final stage would be checked.
23. Checking of DC Output of the DAVR.
24. Offline checking of all limiter circuit is to be done as per OEM procedures.
25. Restoration of parameters altered during testing and re-verification all tunes/parameters.
26. All protections and interlocks would be checked.
27. AVR room temperature (not more than 24 deg C) & humidity (less than 55% RH) control, sealing and cleanliness shall be ensured. Alarm for the same is to be provided in control room for stations where excitation control room is separate

4.4 DYNAMIC TESTS

a) Part-1:

Following tests will be carried out at Turbine speed 3000 rpm:

1. Measurement of the PMG output voltage and phase sequence.
2. Measurement of PMG Currents.
3. Carrying out open circuit characteristics (OCC) and Short circuit characteristics (SCC) tests (Details Ref – Annexure – V).
4. Check the thyristor current and mV drop across thyristors.

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b) Part-II:

1. After carrying out of SCC/OCC Tests and normalization of all settings, before going for synchronization Change over from Auto-1 to Auto-2 and vice versa would be checked.

c) Part – III:

1. PSS Response test needs to be carried out at low load(280-300MW)
2. PSS Response test needs to be carried out at Full Load (325-375 MW)
3. PSS Response test needs to be carried out at intermediate load(400-450MW)

d) Part –IV: Step test on load

1. Step test (+/- 3%) is to be done on different loads
2. Step test needs to be carried out at low load(280-300MW)
3. Step test needs to be carried out at Full Load (325-375 MW)
4. Step test needs to be carried out at intermediate load(400-450MW)

e) Part-V: Limiter checking on load

1. Low excitation limiter
2. Rotor current limiter

5.0 FREQUENCY OF CHECKING AND MEASUREMENTS

Tests shall be carried out during unitoverhaul/recommissioning.

6.0 RESPONSIBILITY CENTRE FOR IMPLEMENTATION

Basic responsibility lies with respective Area in-charge. Overall responsibility lieswith AGM (EM).

7.0 EXCEPTION REPORTING & DISPENSATION OF DEVIATION

Any deviation shall be reported to HOD (EM) and can be dispensed with hisapproval.

8.0 REVIEW

Documents will be reviewed once in two years by HOP.

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9.0 CHECKLIST

Annexure-I

Excitation Test Report

Parameter : U_g

Voltage injection - Channel-1 [$\text{Ratio } 21000/\sqrt{3}/110//\sqrt{3}V$]

Phase	Injected voltage, V	Uac, V	Udc, V	%at Micro Terminal
RS				
ST				
TR				

Voltage injection - Channel-2 [$\text{Ratio } 21000/\sqrt{3}/110//\sqrt{3}V$]

Phase	Injected voltage, V	Uac, V	Udc, V	%at Micro Terminal
RS				
ST				
TR				

Annexure-II

Parameter : I_g

Current injection - Channel-1 [$\text{Ratio } 21000/5A$]

Phase	Injected current	Uac, V	Udc, V	%at Micro Terminal
R				
S				
T				

Check the effectiveness of the CT shorting links.

Current injection - Channel-2 [$\text{Ratio } 21000/5A$]

Phase	Injected current	Uac, V	Udc, V	%at CT Terminal
R				
S				

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T				
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Check the effectiveness of the CT shorting links.

Annexure – III

Parameter - If

Current injection - Channel-1 [Ratio 100/1A]

CT Terminal	Phase	Injected current	Udc, V	% at micro terminal
At CT secondary	R			
At CT secondary	T			

Current injection - Channel-2 [Ratio 100/1A]

CT Terminal	Phase	Injected current	Udc, V	% at micro terminal
At CT secondary	R			
At CT secondary	T			

Annexure – IV

Light Load Test

1. Channel # 1

- Connect 3 phase 220V, 50Hz to the input of the PMG input terminals.
- Connect 5A rheostat to the output (field cable to the exciter is removed and load connected).
- Switch S 12 in service.
- Remove the card UN0663. Change the switch position S865 to 1 (for External U_c).
- Close field breaker and Switch ON Excitation.
- Channel # 1 is in Service. Channel # 2 is minimum,.
- Increase the value of U_c from -ve to +ve and note down the variation of DC voltage and current at the output of the bridge.

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Input voltage – 220V

	U_c	DC Voltage, V	DC Current, A
1	-8.36		
2	0.0		
3			
4			
5			
6			
7			
8			
9			

2. Repeat the same with channel # 2.

- Close field breaker and Switch ON Excitation.
- Channel # 2 is in Service. Channel # 1 is minimum.
- Increase the value of U_c from -ve to +ve and note down the variation of DC voltage and current at the output of the bridge.

Input Voltage – 220V

	U_c	DC Voltage, V	DC Current, A
1	-8.36		
2	0.0		
3			
4			
5			
6			
7			
8			
9			

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Annexure – V

GENERATOR SCC

Sl No.	Ig%	Uc (Vdc)	If %	Exe. Voltage (VDC)	Exe current (A)	Panel KAmp	Panel Rotor Current (A)	Gen KV

GENERATOR OCC

Sl no.	Ug%	Uc	If %	Exe voltage	Exe current	Panel KV	Panel Rotor Current (A)

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ANNEXURE VI**LIMITER CHECKING OFF LINE FORMATS****SIMULATION:**

- Disconnect DC o/p of DVAR
- Simulate Speed > 90%
- Remove Grp prot
- Reset ULR
- Remove both uno663 card & change the switch position S865/1 to 1 position for external Uc in both channels.
- Open terminals links for Voltage for Channel 1&2.
- Current injection ch-1:Open current links for channel 1
- Current injection ch-2:Open current links for channel 2
- Note the alarms & led before testing.

V/F LIMITER:

Calculation

F280=110 %

V=110, f=50

 $V/f = 110/50 = 2.2 = 100\% \times (\text{setting}) = 110\%$ $X = 2.2 \times 110\% = 2.42$

Varying frequency	Varying voltage
110/f=2.42	V/50=2.42
F=45.45hz (limiter frequency operation value)	V=121v (limiter voltage operation value)

Channel - 1

Channel 1 in service and uc -ve.

Field breaker on & excitation on.

Inject voltage 110v a.c in both channels simultaneously (63.5v at kit).

Match Uref with Uact by raising the Uref.

Decrease the frequency gradually.

Limiter operation

<i>Set</i>	
<i>Reset</i>	

<i>Alarm DAVR</i>	
<i>DDC</i>	
<i>ECD</i>	

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Channel - 2*Channel 2 in service and Uc -ve.**Field breaker on & excitation on.**Inject voltage in both channels simultaneously (63.5v at kit).**Match Uref with Uact by raising the Uref.**Decrease the frequency gradually.***Limiter operation**

Set	
Reset	

Alarm DAVR	
DDC	
ECD	

Load angle limiter*F4A0= 77 deg***Channel 1***Channel 1 in service and uc -ve**Field breaker on & excitation on**Inject voltage 110v ac in both channels simultaneously (63.5v at kit).**Inject current at CH-1: (2.0amp at angle 20deg)**Match vref with Uact by raising the Uref.**Increase current gradually by 0.1 amps***Limiter operation**

Set	
Reset	

Alarm DAVR	
DDC	
ECD	

Channel 2*Channel 2 in service and uc -ve**Field breaker on & excitation on.**Inject voltage 110v ac in both channels simultaneously (63.5v at kit).**Inject current at CH-2: (2.0amp at angle (-20)deg).**Match vref with Uact by raising the Uref.**Increase current gradually by 0.1 amps.***Limiter operation**

Set	
Reset	

Alarm DAVR	
DDC	
ECD	

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Inductive stator current limiter

F430= 105%, F442=21.6p.u change it to 1p.u (both channel)

Channel 1

Channel 1 in service and uc -ve

Field breaker on & excitation on

Inject voltage 110v ac in both channels simultaneously (63.5v at kit).

Inject current at CH-1: (4.0amp at angle (-)5 deg)

Match vref with Uact by raising the Uref.

Increase current gradually by 0.1 mps

Limiter operation

Set	
Reset	

Alarm DAVR	
DDC	
ECD	

Channel 2

Channel 2 in service and uc -ve

Field breaker on & excitation on

Inject voltage 110v ac in both channels simultaneously (63.5v at kit).

Inject current at CH-2: (4.0amp at angle 5deg).

Match vref with Uact by raising the Uref.

Increase current gradually by 0.1 mps

Limiter operation

Set	
Reset	

Alarm DAVR	
DDC	
ECD	

Capacitive stator current limiter

F460= 105%,

Channel 1

Channel 1 in service and uc -ve

Field breaker on & excitation on

Inject voltage 110v ac in both channels simultaneously (63.5v at kit).

Inject current at CH-1 (4.0amp at angle +5 deg)

Match vref with Uact by raising the Uref.

Increase current gradually by 0.1 Amps

Limiter operation

Set	
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Alarm DAVR	
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Reset	
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DDC	
ECD	

Channel 2

Channel 2 in service and uc -ve

Field breaker on & excitation on

Inject voltage 110v ac in both channels simultaneously (63.5v at kit).

Inject current at CH-2: (4.0amp at angle (-)5deg).

Match vref with Uact by raising the Uref.

Increase current gradually by 0.1 Amps

Limiter operation

Set	
Reset	

Alarm DAVR	
DDC	
ECD	

Field current Maximum limiterF460= 105%, f416=18.7 ~~p.u.~~ change it to 1p.u (both channel)**Channel 1**

Channel 1 in service

Field breaker on & excitation on

Inject current at CH-1(0.7 amp)

Increase current gradually by 0.01 mps

Limiter operation

Set	
Reset	

Alarm DAVR	
DDC	
ECD	

Channel 2

Channel 2 in service

Field breaker on & excitation on

Inject current at CH-2. (0.7 amp)

Increase current gradually by 0.01 mps

Limiter operation

Set	
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DDC	
ECD	

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Field current Minimum limiter

F410= 10%

Channel 1

Channel 1 in service

Field breaker on & excitation on

Inject current at ch-1 (0.7 amp)

Increase current gradually by 0.01 mps

Limiter operation

Set	
Reset	

Alarm DAVR	
DDC	
ECD	

Channel 2

Channel 2 in service

Field breaker on & excitation on

Inject current at ES: Ch-2 (0.7 amp)

Increase current gradually by 0.01 mps

Limiter operation

Set	
Reset	

Alarm DAVR	
DDC	
ECD	

Restoration:

- Connect DC o/p of DVAR
- Normalize Speed >90%
- Normalize Grp prot
- Normalize ULR
- Remove both uno663 card & change the switch position S865/1 to 0 position in both channels.
- Close terminals links of Voltage: link ch1 & ch2
- Close Current links: ch-1
- Close Current links: ch-2
- Reset all alarms & led in DAVR.

TESTED BY

CHECKED BY